

Element A

The komodo dragon exhibit at the Nashville Zoo is one of the biggest komodo dragon exhibits in America. However, the zookeepers, James and Cayden, have expressed that the komodo dragons prefer to sit still rather than move around and exercise. The zookeepers want an enrichment toy that can help the komodo dragons move around more for better health, especially as two of the komodo dragons at the exhibit needed special medical attention due to their age. Because the komodo dragons have incredibly sharp, iron-coated teeth and a rather strong bite, it is difficult to build an enrichment toy that can withstand their violent attacks. The zookeepers mentioned that they prefer a pulley system, where when in use, the zookeepers would pull on the wire to move the food across the exhibit. This will result in the komodo dragons chasing after the food and getting them to exercise. Our primary stakeholders are the komodo dragons themselves (Lil Sebastian, Olivia, and Agnes, along with the two babies), as they are the primary users of the enrichment toy. For this reason, the enrichment toy should effectively make the komodo dragons move around more, and it should also entertain them when they are fed from every week to every month. Another set of primary stakeholders are James and Cayden, since they are the ones operating the toy for the dragons' use. Thus, the enrichment toy should be easily operable by the zookeepers. Another primary stakeholder is the outdoor enclosure that houses the komodo dragons, as our solution will be put in the enclosure and could negatively affect it. Ideally, our solution should not cause any damage to the enclosure and should be easily removed, or look natural to their habitat. Some secondary stakeholders are the patrons of the zoo, as they get to see the komodo dragons in action. The enrichment toy should not cause the people outside of the enclosure to be affected so that the patrons will not leave negative words to the zoo (except for the zookeepers who operate the device). The veterinarians and the zoo owners are also

stakeholders since they will spend less time/money taking care of the komodo dragons' health. The enrichment toy should not cause any harm to the komodo dragons.

Designing a proper enrichment toy would help the komodo dragons move around more and ultimately improve their health. It could also potentially make the zoo visitors more excited about the exhibit and more likely to visit again.



Additionally, here is a picture of the exhibit that the komodo dragons will be in when outside.



London, King's College. "Komodo Dragons Have Iron-Coated Teeth to Rip Apart Their Prey." King's College London, 24 July 2024, www.kcl.ac.uk/news/komodo-dragons-have-iron-coated-teeth-to-rip-apart-their-prey.

"Keeping up with the Komodos." Nashvillezoo.org, 2026, nashvillezoo.org/our-blog/posts/keeping-up-with-the-komodos. Accessed 24 Feb. 2026.

Smithsonian's National Zoo & Conservation Biology Institute. "Komodo Dragon." Smithsonian's National Zoo, 9 July 2018, nationalzoo.si.edu/animals/komodo-dragon.

Jirik, Kate. "LibGuides: Komodo Dragon (Varanus Komodoensis) Fact Sheet: Behavior & Ecology."
ielc.libguides.com, 2024, ielc.libguides.com/sdzg/factsheets/komododragon/behavior.





